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Decomposing the composite face effect: Evidence for non-holistic processing based on the ex-Gaussian distribution

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Abstract

Composite faces fuse the top and bottom halves from two different faces to create a powerful illusion of a novel face. It has been argued that composite faces are processed holistically, namely that the constituent face parts are perceived as a template, rather than independent features. This study sought to uncover the locus of the composite face effect by relating its empirical reaction time distributions to theoretical ex-Gaussian parameters. The results showed that the composite face effect for unfamiliar (Experiment 1) and familiar (Experiment 2) faces is generated by pure changes in the exponential component of the ex-Gaussian distribution. This held true for both partial and complete design measures. The exponential component has been attributed to working memory and attentional processes. The results suggest the involvement of attentional and working memory processes in the composite face effect and in the perception of faces in general. They cast doubts on the holistic nature of face processing. The results also provide important constraints on future computational theories of the effect.

Keywords

Composite faces; ex-Gaussian distribution

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Introduction

Many researchers assume that faces are processed holistically. According to the holistic account, face parts or features are perceived together as a single template rather than a collection of unrelated and independent features (Diamond & Carey, 1986; Farah et al., 1998; Maurer et al., 2002; Rossion, 2013; Tanaka & Farah, 1993). One of the most compelling sources of evidence for holistic processing comes from composite faces. These stimuli fuse the top and bottom halves from two different identities to give the impression of a novel face (Young et al., 1987). The illusion is reduced or gone when the face halves are misaligned or inverted (Fitousi, 2016, 2019; Richler, Gauthier, et al., 2008). The illusion has been documented with familiar faces (Young et al., 1987) and unfamiliar faces (Fitousi, 2013, 2015; Hole, 1994; Michel et al., 2006; Richler, Tanaka, et al., 2008) alike. The composite face task has become a standard routine in assessing sundry aspects of face recognition (DeGutis, Wilmer, et al., 2013; Rezlescu et al., 2017).

However, despite extensive research, the psychological mechanisms that generate the composite face illusion are still not understood. Also, little is known about the relations between holistic and attentional mechanisms. Moreover, the popular concepts of "holistic" or "configural" processing have been recently challenged (Burton et al., 2015; Fitousi, 2013, 2015, 2016, 2019b). And their operational definitions have become a matter of debate (Richler & Gauthier, 2014; Rossion, 2013). In addition, researchers have started questioning the idea that familiar and unfamiliar faces are governed by the same holistic mechanisms (Burton et al., 2015; but see Abudarham et al., 2019). The lack of consensus among researchers over these basic issues impedes progress in the field of face perception (Burton, 2013). This begs the applications of novel and fine-grained methods that will allow us to sharpen our hypotheses and construct stronger tests of rudimentary concepts.

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The goal of this study is to gain new insights into the perception of faces by taking advantage of the information lurking in entire reaction time (RT) distribution (Luce, 1986; Van Zandt, 2000). To do so, the present investigation focused on the speeded versions of the composite task with unfamiliar (Experiment 1) and familiar (Experiment 2) faces. The aim here was to discover statistical properties that characterise the entire distributions of RTs. A second goal was to test new hypotheses based on these regularities. The parameters of a well-known theoretical distribution—the *ex-Gaussian* (Burbeck & Luce, 1982; Hohle, 1965)—were fit to the empirical data, and then related to experimental manipulations (Heathcote et al., 1991). This tactic has proved successful in other areas of cognitive research (Buzy et al., 2009; Gordon & Carson, 1990; Leth-Steensen et al., 2000; Rohrer, 1996; Spieler et al., 1996). In the current effort, such analyses culminated in the discovery of statistical effects in the perception of composite faces that allowed us to uncover mechanisms that govern the perception of composite faces and faces in general.

The composite face illusion

The composite face illusion has been invented 30 years ago in a seminal study by Young et al. (1987). In the original paper, top and bottom halves from two familiar faces (e.g., Mick Jager, David Bowie) were fused together to create a novel face. Observers experienced the composite as a new face, an impression that was sufficiently powerful to interfere with the identification of the original face parts. Young et al. compared performance in aligned and misaligned composites and found that participants were slower and more error prone to name the top half when the two halves were aligned than when they were misaligned. The effect was abolished with inverted faces. Hole (1994) has extended this paradigm to unfamiliar faces using a delayed matching task, in which participants were presented with a study and then a test face. The participants' task was to decide whether the top half of the test face is "same" or "different" from that of the study face. In this paradigm, it is more difficult to judge whether the top halves of two faces are "same" or "different" when the two halves are aligned with "different" bottom halves than when they are misaligned. Researchers have routinely adopted this delayed matching task, using two allegedly incompatible measures of the composite face effect. Both measures can be derived from the same experiment. The *complete design* measure is computed as a Congruency × Alignment interaction term (Richler & Gauthier, 2014), and gauges better performance in congruent (i.e., both top and bottom parts are "same" or both are "different") than in incongruent (i.e., top is "same" and bottom is "different" or vice versa) faces; this congruency effect is decreased or abolished with misaligned faces. The *partial design* measure (Rossion, 2013) is computed as a difference in performance between aligned and misaligned conditions only for *same* trials. This measure is considered to reflect the nature of the perceptual illusion because appearance of one (top) half of a face does not look like itself when conjoined with another (bottom) half.

Proponents of the complete and partial effects have argued that the two measures gauge different mechanisms. In support of this claim, Richler and Gauthier (2014) have presented evidence for low and insignificant correlations between the two effects. However, these researchers based their conclusions only on the signal detection theory (SDT) indices. When Fitousi (2019b) measured the correlation in terms of mean RTs (MRTs) and accuracy rates, he found strong (in accuracy) and medium (in RT) significant correlations. Fitousi's (2019b) results support the relatedness of the two types of measures, and also raise further questions regarding the nature of the (holistic?) mechanisms that derive the two measures. The notion of holistic processing as a unique characteristic of faces is vague and may impede progress in research on faces (Burton, 2013). This study expands the scope of theoretical possibilities by relating face processing to more general mechanisms such as attention and working memory.

Holistic accounts of the composite face effect

What does "holistic face processing" mean? Maurer et al. (2002) have proposed a widely adopted distinction between three types of holistic processing: (a) the perception of "first-order" relations, which refers to the basic arrangement of the facial features (e.g., eyes appear above a nose); (b) the perception of a face as a Gestalt or template; and (c) the sensitivity to second-order relations, such as the spatial distances between features. Because the alignment manipulation in the composite face design likely affects the overall template, the effect has been often considered as a manifestation of definition (b), namely, the resultant of a Gestalt perception (Burton et al., 2015; Rossion, 2013). However, because the composite face effect is often observed as a congruency effect with unfamiliar faces in a delayed matching task (Richler & Gauthier, 2014), or as an alignment effect, in which a face half (top) is compared with a previously seen one in aligned and misaligned configuration (Rossion, 2013), it also fits into definition (c), namely sensitivity to second order relations.

The two main types of composite face effects described earlier are the most popular in the literature. And proponents of each measure adhere to their preferred mechanism of holistic processing.[1] Of course, this should not be necessarily so, as the accounts are independent of the measures. Exponents of the partial design propose a template hypothesis (Rossion, 2013), according to which faces are encoded initially as a template or a "blob" where the individual features are not accessible (Farah et al., 1998; Tanaka & Farah, 1993; Young et al., 1987). This

coarse representation becomes increasingly detailed as featural information becomes accessible to perception (Sergent, 1986; see also Jacques & Rossion, 2009). The alignment manipulation disrupts the encoding of template and allows part-based (featural) perception to dominate. The template hypothesis is characterised by specific predictions regarding the temporal evolution of the effect, predicting a strong effect at early processing stages, and weaker effect at later stages, as the influence of part-based information progressively increases (Sergent, 1986).

An alternative explanation of the template hypothesis is an attentional hypothesis, which is often brought up by proponents of the complete design measure (Richler et al., 2011, 2012). According to this approach, holistic processing results from "an attentional strategy that has become automized with experience" (Richler & Gauthier, 2014, p. 1282). Faces are initially encoded and represented as a collection of independent face-parts, but "holistic processing arises from obligatory encoding of all object parts because a strategy of attending to all parts cannot be 'turned off'" (p. 1282). According to this idea, attention to the relevant (top) part fails due to the massive amount of experience observers have with faces (Gauthier et al., 1998). Although proponents of the attentional account claim that it is unique to faces (due to enormous experience with these stimuli), they overlook the fact that such an account bears close affinity to a more general mechanism known as object-based attention (Duncan, 1984; Kahneman & Henik, 1981). According to object-based attention views, once an object has been attended, all its parts are activated. It is difficult to see how faces are different from other objects in this respect.

Adjudication between the attentional and perceptual accounts has proved particularly challenging due to the lack of concrete processing models and exact predictions (Fitousi, 2015, 2016). Moreover, the selective attention conception of holistic processing (Richler & Gauthier, 2014) has been challenged by the proposal that the complete design effect, which is sometimes called a congruency effect (Rossion, 2013), probes the same selective attention and control mechanisms that produce the congruency effects in Stroop (1935) and flanker (Eriksen & Eriksen, 1974) paradigms (Fitousi, 2015; Rossion, 2013; Von Der Heide et al., 2018).

Unfortunately, there is no psychologically and computationally plausible mechanism that can explain how the alignment or congruency manipulations relate to cognitive operations. Proponents of both the attentional (Richler & Gauthier, 2014) and template (Rossion, 2013) approaches assume that the interference or alignment effects are produced by some sort of integration of information from the two face halves. The template hypothesis states that integration occurs due to an initial holistic encoding of the features (Jacques & Rossion, 2009; Sergent, 1986). The attentional approach postulates that integration happens due to lifelong experience with the complete face (Richler, Palmeri, Gauthier, & Wenger, 2008). However, both rely on operational definitions that are susceptible to circularity and lack of consistency (Richler, Gauthier, Wenger & Palmeri, 2008).

Non-holistic accounts of the composite face effect

The emergence of a composite face effect does not necessarily imply that the information from the two face halves has been integrated. The effects can occur due to differences in relative discriminability between the top and bottom parts (Fitousi, 2016; Fitousi & Algom, 2006; Melara & Mounts, 1993), or due to decisional factors across conditions (Fitousi, 2016; Richler et al., 2011; Von Der Heide et al., 2018), which are only remotely related to holistic processing (Townsend et al., 2018). Second, when tested in related paradigms or against computational models of holism, the integration (holistic) hypothesis is not well supported (Bradshaw & Wallace, 1971; Donnelly et al., 2012; Ellison & Massaro, 1997; Fitousi, 2015, 2016; Gold et al., 2012; Loftus et al., 2004; Macho & Leder, 1998; Sergent, 1984; Tversky & Krantz, 1969; Wenger & Ingvalson, 2002, 2003). For example, Fitousi (2015) deployed the Garner paradigm (Garner, 1974; for review see Algom & Fitousi, 2016)—a classic measure of holistic processing—to test the integrality of top and bottom halves in composite faces. He found no interference between the two components, a result that supported their separability (see also Amishav & Kimchi, 2010; Pomerantz et al., 2003; Richler et al., 2013).

Strong evidence for the absence of information integration in composite faces comes from application of the *redundant target task* (Miller, 1982) and the *system factorial technology* (SFT, Townsend & Nozawa, 1995) to composite faces (Fitousi, 2015; see also Cheng, McCarthy, Wang, Palmeri & Little, 2018). The SFT is a powerful mathematical theory accompanied by a set of methodological tools capable of diagnosing fundamental processing characteristics (Fitousi, 2019a; Fitousi & Algom, 2018, 2019). The SFT affords qualitative and quantitative identification of the underlying processing system (serial, parallel, or coactive). The question is whether the information from top and bottom parts is processed one after the other (serially), simultaneously (parallel), or integrated such that evidence from the two parts coalesce into a single channel (coactive). Both Fitousi (2015) and Cheng et al. (2018) found little evidence for a coactive system and hence for true integration of information. Ample evidence was adduced to support serial or parallel systems. These findings cast serious doubts on the holistic idea of information integration (Fitousi, 2015, 2016, 2018).

Recently, Burton et al. (2015) have also criticised the deployment of holistic or configural (second-order relations between facial features) accounts to explain familiar faces recognition. Their arguments revolved around four

central points: (a) the notion of configurality is underspecified, (b) large configural changes leaves face recognition unharmed (Bindemann et al., 2008), (c) recognition is impaired by nonconfigural changes (Rhodes, 1996), and (d) identification is dominated by texture rather than shape (Hill et al., 1995). Burton et al. were careful to distinguish between holistic (template) and configural (second-order relations) processing, arguing that holistic (template) processing is still a dominant strategy in the composite face effect with familiar faces. However, given that the composite face effect with unfamiliar faces is based on a delayed matching task in which the configurality (second-order relations) is also changing from study to test, Burton et al.'s criticism renders the holistic account of the composite face effect suspect.

Fitousi (2015, 2016) surmised that the composite face effect is not unique to faces but reflect a general attentional strategy akin to *object-based attention* (Duncan, 1984; Kahneman & Henik, 1981). In object-based attention, all constituent features or parts are activated once an object has been selected. This type of account can explain, for example, the decrease in the size of the Stroop conflict when colour and word are spatially separated (Fitousi & Algom, 2006; Kahneman & Chajczyk, 1983). The reduction in the congruity composite effect with misalignment is likely due to a similar disruption of objecthood (Fitousi, 2015).

Does the composite face effect engage attentional mechanisms?

Rossion (2013) has claimed that the complete design measure does not capture holistic processing but rather probe processes of selective attention and conflict that are studied by the Stroop or flanker effects. To address this possibility, Gauthier et al. (2018) have used an individual-differences approach to study the relations between the complete design (congruency) measure of the composite face illusion and several versions of the Stroop and Flanker tasks. They have collected data from 130 participants who completed a non-speeded version of the composite task and speeded versions of the Stroop and Flanker tasks, as well as fluid IQ measurement. Gauthier et al. did not find correlations between the magnitude of the composite effect and the size of the Stroop and Flanker effects. They also did not find correlation between the composite effect and scores of fluid IQ. They did detect a small correlation between the Stroop and Flanker effects. The authors interpreted these results as suggesting that the complete design measure does not tap the same attentional processes captured by the Stroop and Flanker effects. However, this study is problematic for several reasons. First, it deploys a non-speeded version of the composite effect and correlates it with the speeded measures of the traditional selective attention tasks. A more appropriate measure of relations between the two should include scores of the speeded version of the composite task. Second, this study does not replicate a great number of previous studies who did record high and significant correlations between fluid IQ and congruency effects in the Stroop and Flanker tasks (Schmiedek et al., 2007; Tse et al., 2010). The upshot is that more work should be held to uncover the role of attention and control in the composite face effect.

Our review so far brings forward three major issues that require further scrutiny: (a) the working definition of holistic processing, (b) its role in the recognition of familiar and unfamiliar faces, and (c) its relations to other conflict tasks. A useful goal is therefore to start posing tighter constrains on alternative theories of face processing. One way of achieving constraint is through the discovery of regularities in the empirical data. Distributional analyses on entire RT data are excellent method for discovering such regularities (Balota & Yap, 2011; Heathcote et al., 1991; Hockley, 1984). These methods go well beyond the one-dimensionality of the RT mean into the multidimensionality of the RT distributional shape.

Distributional analyses and the ex-Gaussian distribution

Balota and Yap (2011) have noted that "we all love the mean!" (p. 160). This observation is particularly true with respect to face researchers, who have been mainly preoccupied with mean response times (MRTs) or mean error rates when measuring performance with faces, neglecting the important information lurking in the distributional shapes themselves. A powerful approach to understanding the influence of manipulations on RT distributions is to fit a mathematical function to these RT distributions (Luce, 1986). The ex-Gaussian distribution has been a successful one (Heathcote et al., 1991; Ratcliff & Mrdock, 1976). It is a convolution of a Gaussian and an exponential distribution. The probability distribution function (pdf) of the function is expressed as

$$f\left(t \mid \mu,\sigma,\tau\right) = \frac{1}{\tau\left(2\Pi\right)^{\frac{1}{2}}}\exp\left[\frac{\sigma^2}{2\tau^2} - \frac{\left(t-\mu\right)}{\tau}\right] \times \int_{-\infty}^{\frac{(t-\mu)}{\sigma}-\left(\frac{\sigma}{\tau}\right)} \exp\left(-\frac{y^2}{2}\right)\mathrm{d}y$$

with μ (mu) and σ (sigma) serving as the mode and standard deviation of the Gaussian distribution, and τ (tau) as the mode and standard deviation of the exponential distribution (see Figure 1). The overall mean of the distribution is equal to $\mu + \tau$, and the variance is equal to $\sigma^2 + \tau^2$. The ex-Gaussian distribution exhibits excellent fit to RT distributions (Luce, 1986). The main goal of researchers

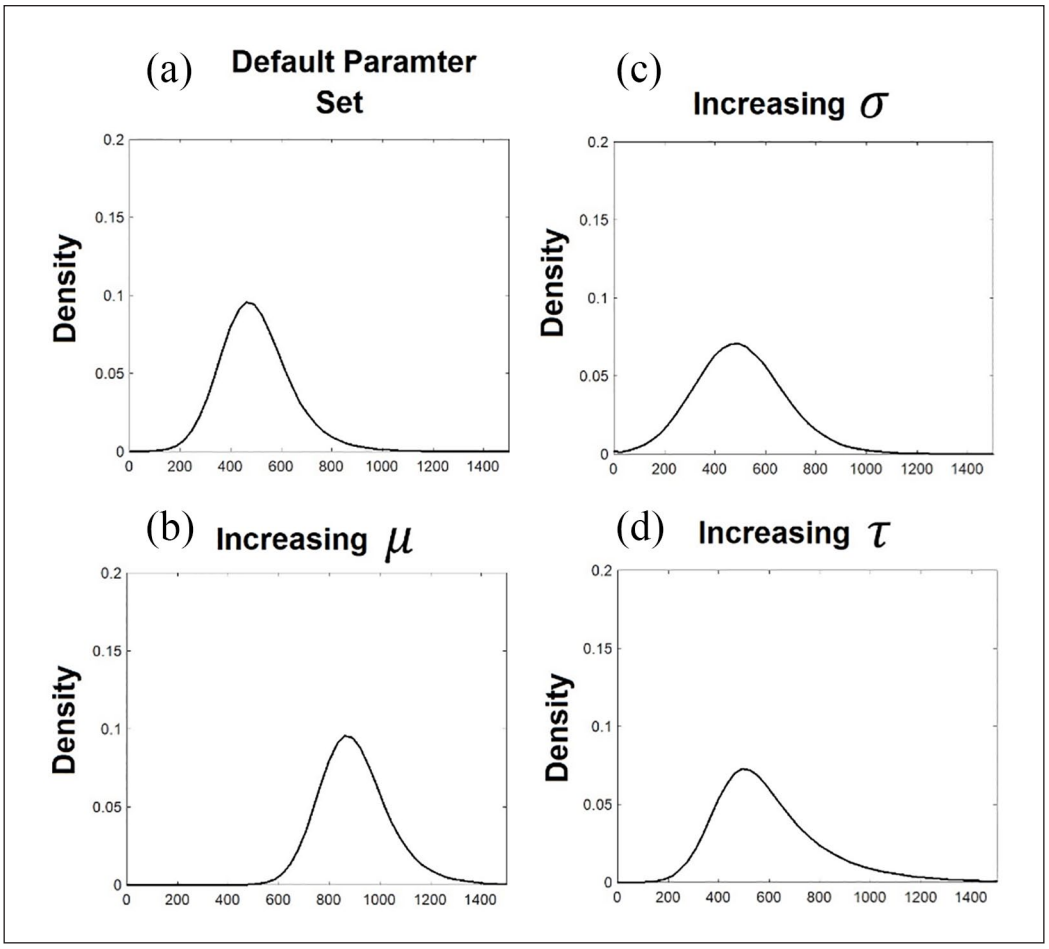


Figure 1. The shape of the ex-Gaussian distribution as a function of changes in the ex-Gaussian parameters. (a) $\mu = 400$, $\sigma = 100$, $\tau = 100$ (default parameters), (b) $\mu = 800$, $\sigma = 100$, $\tau = 100$ (increasing μ), (c) $\mu = 400$, $\sigma = 150$, $\tau = 100$ (increasing σ), and (d) $\mu = 400$, $\sigma = 100$, $\tau = 200$ (increasing τ).

applying this approach is to discover which of the three parameters are affected by manipulations and experimental conditions (Balota & Yap, 2011). The approach has been applied successfully to various questions in such diverse areas as word reading (Balota et al., 2008), individual differences in the Stroop task (Heathcote et al., 1991; Spieler et al., 1996), or memory (Hockley, 1984).

The ex-Gaussian parameters: empirical and theoretical meanings

The ex-Gaussian analyses have been applied to many research questions (Balota & Yap, 2011). The most relevant area of application for the current purpose is attention research, where this methodology assisted in dissociating different types of tasks, stimuli, and attentional mechanisms in the Stroop, Flanker, and Navon tasks (Spieler, Balota, & Faust 2000). The ex-Gaussian analyses can also be used to understand the commonalities and discords between the various species of composite effects. For example, if the partial and complete design effects surface in different ex-Gaussian parameters, they are likely governed by different mechanisms[2] (see Schwarz & Miller, 2012). A similar logic applies to the emergence of composite effects with familiar and unfamiliar faces. The ex-Gaussian analysis can also shed light on the locus of the effect and its temporal dynamics. Effects on τ reflect unique influence on the tail of the distribution, and therefore on later percentiles (Balota & Yap, 2011; Fitousi, 2020; Roelofs, 2012), suggesting that the effect is increasing with time. Effects on μ entail that the size of the effect remains constant across time.

Another benefit of the ex-Gaussian analysis is the ability to relate the ex-Gaussian parameters to cognitive mechanisms. The Gaussian and exponential parameters have been related to opposing cognitive processes. Balota and Spieler (1999) ascribed the Gaussian parameter μ to "more stimulus driven automatic (nonanalytic) processes" and the exponential component τ to a more central attention demanding (analytic) processes (p. 348). This maxim might be too general, as the mapping between the

ex-Gaussian parameters and cognitive mechanisms is not straightforward (Fitousi, 2020; Heathcote et al., 1991; Matzke & Wagenmakers, 2009; Rieger & Miller, 2019). However, recent studies offer a set of strong links between ex-Gaussian parameters and cognitive processes in traditional attention paradigms such as the Stroop, Flanker, and Navon tasks. Although the pertinent hypotheses are still under scrutiny (Roelofs, 2012), they have gained traction, and their close affinity to the paradigmatic measures deployed in the face literature affords the starting of a principled hypotheses testing.

There are three main hypotheses concerning the involvement of ex-Gaussian components in attentional mechanisms and processes. These hypotheses have been rigorously studied in the field of attention (Roelofs, 2012; Schmiedek et al., 2007; Steinhauser & Hübner, 2009). The basic idea is that unique cognitive operations selectively influence exclusive ex-Gaussian parameters, and therefore influence the shape of the RT distribution in a specific manner. Incidentally, all three hypotheses attribute a central role to the contribution of the τ component: (a) τ is related to working memory, control, and selective attention (Schmiedek et al., 2007), (b) τ reflects task conflict (Steinhauser & Hübner, 2009), and (c) τ is affected by the operation of object-based attention (Spieler et al., 2000). Next, I expand on these hypotheses and review the evidence.

Tau and task conflict

A seminal study by Heathcote et al. (1991) demonstrated the utility of the ex-Gaussian in the study of attention. Heathcote et al. used ex-Gaussian analyses in the Stroop colour-naming task. They found that the Stroop effect in mean RTs (MRT) was different from the Stroop effect in the three ex-Gaussian parameters. Participants were asked to vocally name the ink colour of congruent or incongruent colour words (e.g., the word BLUE printed in *green*) or a neutral series of Xs (e.g., XXXXX in *green*). In this design, a global Stroop effect (MRT Incongruent – MRT congruent) consists of two subcomponents: an *interference* (MRT Incongruent – MRT neutral) and a *facilitation* (MRT neutral – MRT congruent). Analyses on MRT revealed a global Stroop effect and an interference, but no facilitation. In contrast, analyses on the ex-Gaussian parameters revealed that in μ there were interference and facilitation (and a global Stroop effect), in τ there was no global Stroop effect, but a reversed facilitation (congruent was slower than neutral), and a regular interference. Hence, in the ex-Gauss analyses, the congruent condition exhibited both facilitation in μ and interference in τ. This was not reflected in the MRT because the MRT is equal to the sum of μ and τ, and the facilitation and interference were of the comparable magnitude. Other studies have confirmed this pattern (Mewhort et al., 1992; Spieler et al., 1996, 2000)

Subsequent studies have shown that the τ component is a strong indicator of task conflict in the Stroop task (Steinhauser & Hübner, 2009). Task conflict refers to the opposing demands posed by a stimulus on viewers. In the Stroop task, both congruent and incongruent stimuli convey opposing tasks—reading (the word) and naming (the colour). In contrast, neutral stimuli (e.g., a series of triangles) do not create task conflict because they do not activate the reading task, but only the colour naming task. Hence, the former type of stimuli is dubbed *bivalent* whereas the latter is called *univalent*. Conflict task results in faster RTs in the neutral than in the congruent condition and specific effects on the τ component (see Heathcote et al., 1991). The possibility that a task conflict exists in the composite task seems unlikely. Although, one can view responses to the two faces halves as two separate tasks.

Tau and working memory

The exponential component τ has been found to be strongly related to working memory capacity and control (Schmiedek et al., 2007; Tse et al., 2010), whereas the Gaussian component has not. Schmiedek et al. (2007) examined individual differences in the magnitude of the three ex-Gaussian parameters. They applied structural equation modelling to a battery of choice RT tasks (CRTs) which probe working memory and fluid intelligence. They found that individual differences in τ, but not in μ and σ, were correlated with individual differences in working memory. Using a similar individual-differences approach, Tse et al. (2010) compared healthy participants with participants at the early stages of dementia (dementia of the Alzheimer type [DAT]). They found that the performance of the DAT patients was correlated with increase in τ. Moreover, τ was also correlated with psychometric measures of episodic/semantic memory, working memory, and processing speed. Balota and Yap (2011) argued that these relations are also supported by the finding that participant's slow RTs are more strongly related to memory performance than are their fast RTs (Coyle, 2003).

Now, how can changes in τ be related to both task conflict and working memory? Schmiedek et al. (2007) and Wilhelm and Oberauer (2006) argued that the better ability to map stimulus and response in working memory results in improved handling of task conflicts. This idea has been supported by the finding that the Stroop interference correlates with intelligence and working memory (Kane & Engle, 2003). Thus, it is likely that effects on working memory can be manifested as effects on task conflict. Unsworth and Spillers (2010) also found reliable relations between τ and measures of working memory and fluid intelligence. They ascribed them to lapses in sustained attention, and particularly to failures to retrieve the task goal from their working memory.

Tau and object-based attention

Spieler et al. (2000) observed that effects on τ appear only when the colour and word are spatially integrated, but not when the two are spatially separated (word and colour appear one above the other). They also found that effects on τ were absent in the Navon local/global task (Navon, 1977), in which observers respond to a large letter (e.g., H or S) composed of small letters (e.g., H from small Ss). Effects on τ were also absent in the Eriksen flanker task (Eriksen & Eriksen, 1974) in which participant respond to a central letter embedded among distractor (e.g., HHSHH). Based on these findings, Spieler et al. (2000) suggested that effects on τ reflect the spatial integration of components, a notion that is consistent with the idea of object-based attention[3] (Duncan, 1984; Kahneman & Henik, 1981). Object-based attention implies the activation of both attended an unattended features or parts of the object. This account is also supported by the decrease in the size of the Stroop conflict when colour and word are spatially separated (Fitousi & Algom, 2006; Kahneman & Chajczyk, 1983). The question that follows is whether a reduction in the composite face effect with alignment is accompanied by an effect on tau?

Predictions

Experimental conditions that are governed by the same processes should exert unique influence on the same ex-Gaussian parameters (Schwarz & Miller, 2012). This logic guided many researchers who applied the ex-Gaussian analysis (Balota et al., 2008), and it can be useful in dissociating between various types of tasks (e.g., Stroop vs. Navon) or different classes of stimuli (e.g., familiar vs. unfamiliar faces). Thus, if the effects of unfamiliar (Experiment 1) and familiar (Experiment 2) composite faces on the ex-Gaussian parameters will not be manifested in the same set of ex-Gaussian parameters, then familiar and unfamiliar faces are likely governed by different processes (Burton et al., 2015). Similarly, if the partial and complete design measures (Experiment 1) influence different ex-Gaussian parameters, then they gauge separate processes/constructs (Richler & Gauthier, 2014). In the opposite case, when two conditions or stimuli affect the same ex-Gaussian parameters, we cannot immediately infer that they are governed by the same processes, but we can increase our belief in the hypothesis that they are dissociable. In addition, if the complete design version of the composite face effect is a species of a Stroop effect, as argued by Rossion (2013) and countered by Gauthier et al. (2018), then the composite effect should influence the same ex-Gaussian components that have been observed in the Stroop task (i.e., τ).

A related set of predictions concern the mechanisms that generate the composite face effect and their relations to the candidate accounts. As mentioned earlier, the Gaussian parameter μ is ascribed to more stimulus driven automatic (nonanalytic) processes, whereas the exponential component τ is related to a more central attention demanding (analytic) processes (Balota & Spieler, 1999). In addition, the abundance of evidence presented here suggests that the τ component in conflict paradigms is related to attention and working memory processes (Schmiedek et al., 2007; Spieler et al., 2000; Steinhauser & Hübner, 2009; Tse et al., 2010). Thus, we can use the ex-Gaussian parameters to separate analytic from holistic processes in the composite effect. According to the template account (Rossion, 2013), the composite face effect reflects a purely holistic, early, stimulus driven process and therefore should be manifested in the Gaussian components (μ and/or σ). According to the object-based attention account (Fitousi, 2015), the composite face effect reflects a purely late, object-based attention process, and therefore should be manifested at the exponential (τ) component. The predictions for the selective attention account (Richler & Gauthier, 2014) are not clear cut, because, on the one hand, this account argues for attentional mechanisms but on the other hand, claims that these mechanisms are different from those involved in classic attentional paradigms (Gauthier et al., 2018). So, if we keep with the former claim, this account is more in accordance with the emergence of the composite face effect in the exponential (τ) component.

Experiment 1

This experiment aimed at replicating the standard partial and complete design composite face effects using the delayed matching task with unfamiliar faces (Hole, 1994). Participants were presented with a study and then a test composite face. The first goal of this experiment is to replicate the standard complete and partial design effects, at the level of the MRTs and accuracy. The second goal is to model the RT distributions with the ex-Gaussian parameters and test the predictions.

Method

Participants. A sample of 20 female undergraduate students were recruited from Ariel University's pool of participants (M_{age}=22.4, SD=2.1). They received course credit for their participation. All participants reported normal or corrected-to-normal vision. All participants gave their written consent for participation.

Stimuli. A total of 40 grey scale faces were cut in half to produce 20 face tops and 20 face bottoms. On every trial, the top and bottom parts of each face were randomly composed. A thin white line separated the face halves. Faces were presented within an oval shape, as grey scale image over black background. In the aligned faces, the two halves were presented exactly one above the other. In the misaligned faces, the top halve was shifted to the middle of bottom half (see Figure 2). Viewed at a fixed distance of 76 cm, the aligned

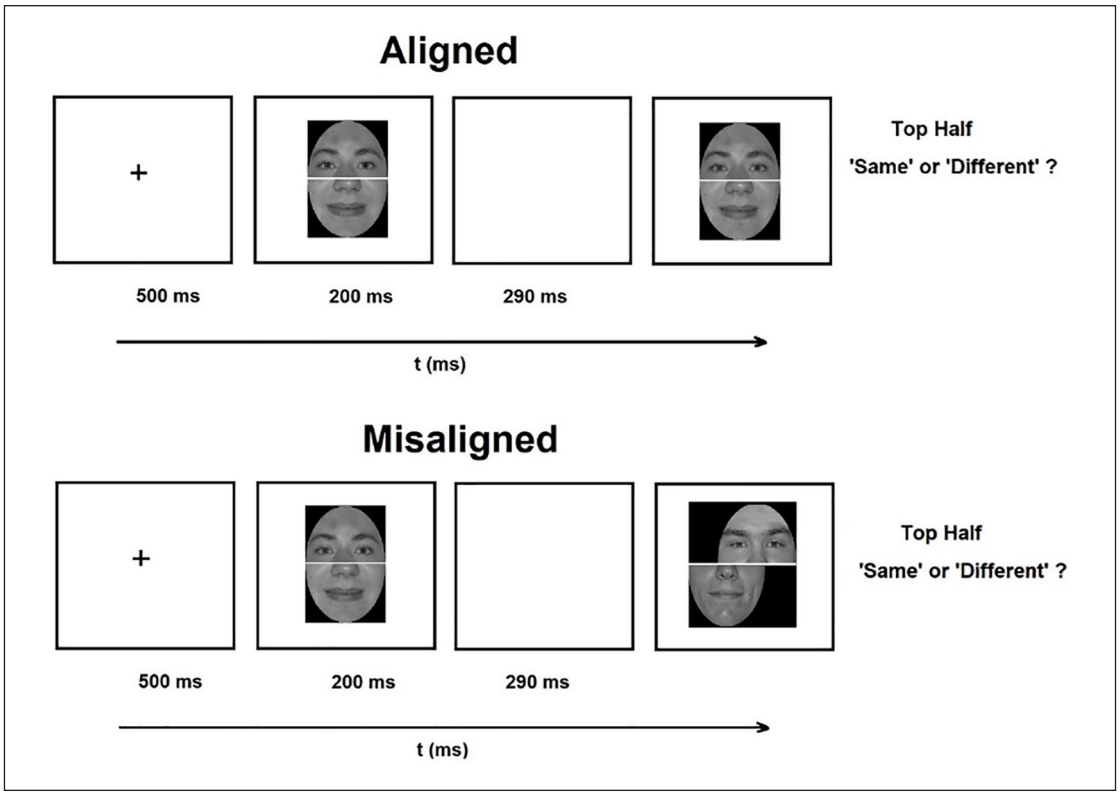


Figure 2. Experiment 1: a typical sequence of an experimental trial.

images subtended 4.4° of visual angle, horizontally, and 2.3° vertically, whereas the misaligned images subtended 6.5° of visual angle, horizontally, and 2.3° vertically.

Procedure and design. Each trial started with a fixation cross for 500 ms, followed by a study face (always aligned) for 200 ms, a blank for 290 ms, and then a test composite face till response. Aligned and misaligned test composite faces were presented in alternated blocks. Each block consisted of 80 trials with all four possible combinations of study and test faces (same-same, same-different, and different-same) equally presented. Each block was repeated six times with the order of stimuli randomly decided by the computer. Overall, each participant completed 960 experimental trials. Participants were instructed to respond as fast and as accurate as they could. The instructions highlighted both speed and accuracy. Participants were asked to press one key ("m") if the top part was "same" with the study face, and another key ("z") if the top half was different from the study face. Response mapping was kept constant for all observers and across conditions (aligned and misaligned) all.

Results

RTs either larger than 2 *SD*s from each participant's mean or smaller than 150 ms were removed from analysis.

Partial design. Figure 3 presents the MRTs and mean error rates in the aligned and misaligned condition for "same" trials only. The partial design effect is a difference in performance measures in the two conditions. The effect was significant for both RT—$t(19)=3.48$, $d=0.23$, $p<.005$—and error rates—$t(19)=8.67$, $d=1.36$, $p<.00005$.

Complete design. Figure 4 presents the MRTs and mean error rates as a function of alignment and congruency level. Analyses on MRTs showed that the effect of alignment was on the verge of significance, $F(1, 19)=3.92$, $\mathrm{MSE}=6{,}190$, $\eta_p^2=0.17$, $p=.06$, entailing faster RTs for misaligned than aligned faces. The effect of congruency, $F(1, 19)=12.9$, $\mathrm{MSE}=7{,}481$, $\eta_p^2=0.40$, $p<.005$, was significant, pointing to faster RTs for congruent than incongruent faces. Most importantly, a significant Congruency × Alignment interaction, $F(1, 19)=10.01$, $\mathrm{MSE}=9{,}107$, $\eta_p^2=0.36$, $p<.01$, reflected a modulation of the congruency effect by alignment. Planned comparison showed that a congruency effect was present in the aligned, $F(1, 19)=12.19$, $\mathrm{MSE}=16{,}548$, $\eta_p^2=0.39$, $p<.005$, but not in the misaligned condition ($F<1$). These results replicate the standard complete design composite face effect (Richler & Gauthier, 2014) in the RT data.

A similar set of analyses was performed on mean error rates. The effect of alignment was significant, $F(1, 19)=66.54$, $\mathrm{MSE}=0.064$, $\eta_p^2=0.76$, $p<.005$, entailing

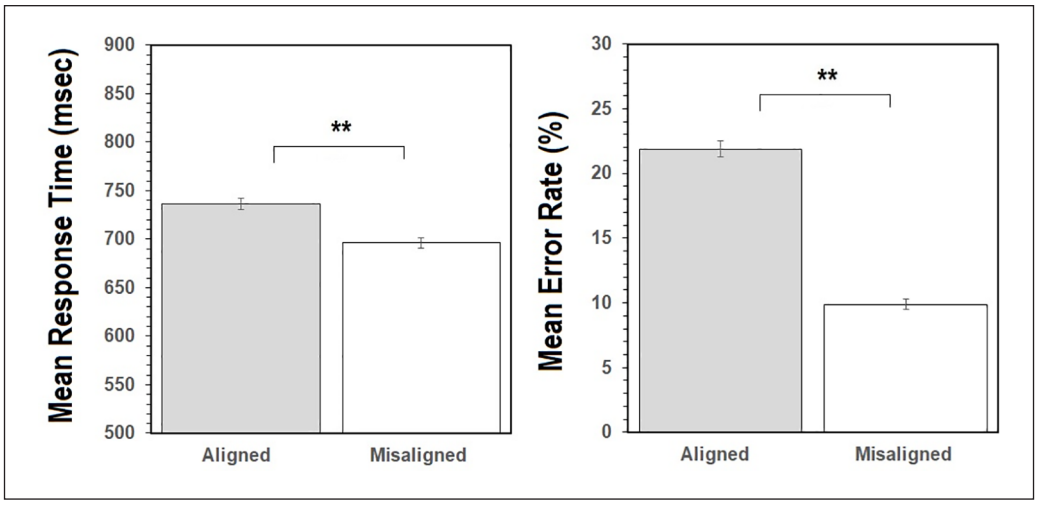


Figure 3. Experiment 1: mean response times (ms) (left panel) and mean error rate (%) in the partial design. Error bars are standard error of the mean.

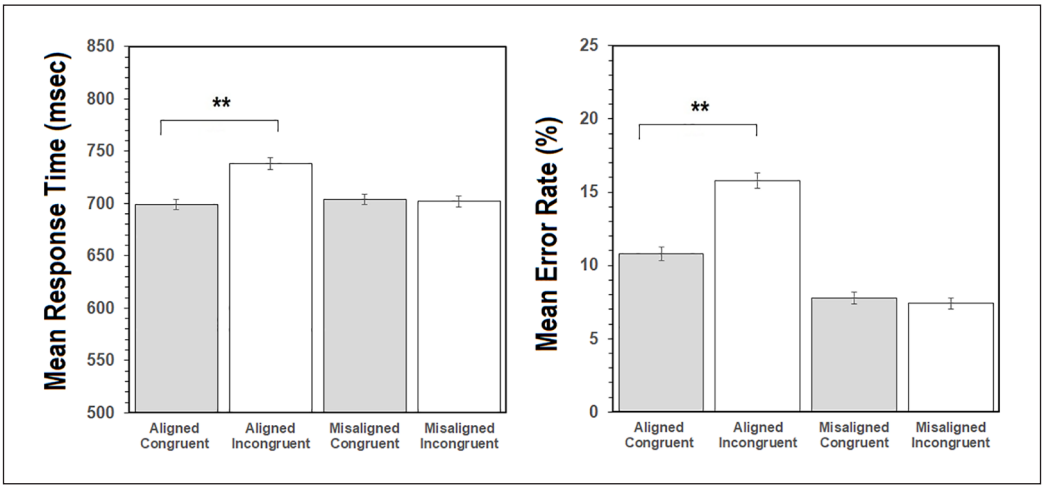


Figure 4. Experiment 1: mean response times (ms) (left panel) and mean error rate (%) in the complete design. Error bars are standard error of the mean.

more errors for aligned than misaligned faces. The congruency effect was significant too, $F(1, 19) = 13.0$, $MSE = 0.010$, $\eta_p^2 = 0.50$, $p < .005$, attesting to lower error rate in the congruent than in the incongruent trials. A Congruency × Alignment interaction comparable with that found with RT, $F(1, 19) = 18.96$, $MSE = 0.014$, $\eta_p^2 = 0.50$, $p < .0005$, was due to the presence of a congruency effect in the aligned condition, $F(1, 19) = 5.94$, $MSE = 0.024$, $\eta_p^2 = 0.11$, $p < .05$, but not in the misaligned condition ($F < 1$).

Fitting the ex-Gaussian distribution

I have deployed two complementary procedures for fitting the ex-Gaussian distribution. The first follows the common practice in the literature of fitting the ex-Gaussian parameters: μ, σ, and τ separately to each of the experimental conditions (Heathcote et al., 1994; Roelofs, 2012). The second is *a model selection approach* (Ashby & Lee, 1991; Fitousi, 2014, 2018) which is more sophisticated, but less prone to inferential pitfalls. It should be noted that in all cases, the two methods of inference converged on the same conclusions. The data in these analyses did not include the error trials, but they were not trimmed in order be able to capture (if exist) effects on the tail (see, Roelofs, 2012).

In the first method, ex-Gaussian parameters are fitted separately to the RT distribution of each experimental conditions of each participant (see Heathcote et al., 1991; Roelofs, 2012). Assessment of fit is achieved by superimposing the estimated ex-Gaussian function on a histogram of the corresponding data and computing the chi-square

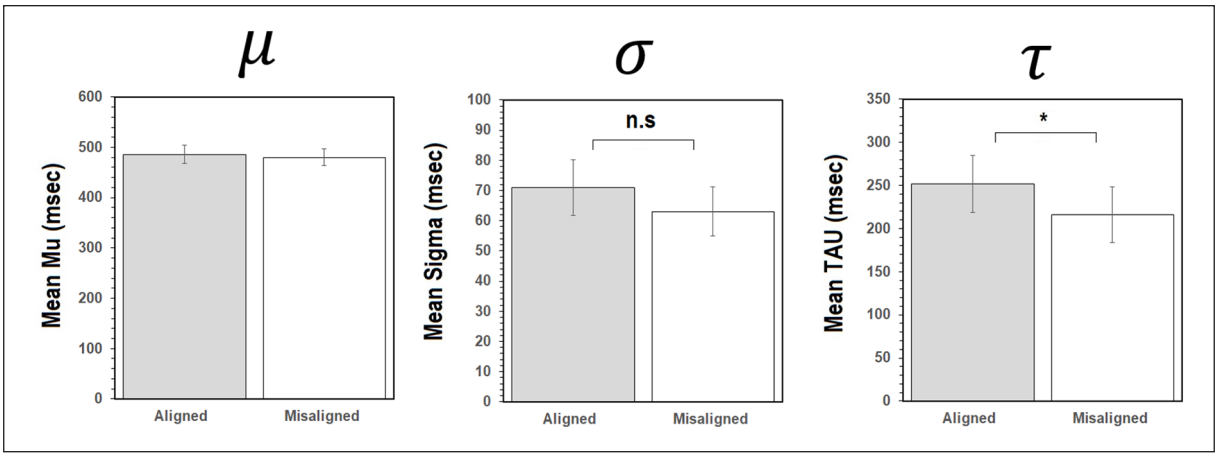


Figure 5. Experiment 1: values of mu, sigma, and tau, as a function of alignment for "same" trials in the partial design measure. Error bars are the standard error of the mean.

(Heathcote et al., 1991). A SIMPLEX optimization algorithm (Nelder & Mead, 1965) minimised a chi-square function. Best-fitting parameters are those that minimised the chi-square function ([Observed-Expected]^2/Expected). The ex-Gaussian function was fitted separately for each participant's correct RT distributions for each experimental condition (Heathcote et al., 1991). This procedure was held twice: once for the complete design and once for the partial design. The former resulted in fitting 80 distributions and the latter in fitting 40 distributions. All fitted values exhibited excellent chi-square log-likelihood values (both exhibiting similar results), as indicated by the alignment of the observed and predicted distributions and the relatively small values of the best-fitting minimisation function (e.g., the chi-square value).

The first method is the most common. However, it may have some problems. The simultaneous fitting of all three ex-Gaussian parameters to each of the RT distributions may lead to biased conclusions. First, estimation of parameters can introduce correlations between estimated values, a problem known as *parameter dependency* (Ratcliff & Tuerlinckx, 2002). This potential pitfall can also inflict the estimation of the ex-Gaussian parameters (Spieler et al., 2000). The mean of the ex-Gaussian is equal to the algebraic sum of τ and μ and to meet this constraint, fitting algorithms may produce dependent or correlated estimations of the two parameters (see also Schmiedek et al., 2007). A second related problem is that distributions could be fit nearly as well with parameter sets that do not vary across conditions. And the difference in the degree of fit between these parameters and the ones that do might not justify preferring the latter over the former.

To overcome these potential inferential problems, I have used a complementary inferential procedure. In this method, models representing competing hypotheses were fitted and compared for degree of fit.[4] This procedure is known as *hierarchical model selection* (Ashby & Lee, 1991; Fitousi, 2014, 2018). The procedure is performed by defining a hierarchy of possible models, beginning with a null model in which μ, τ, and σ are held fixed across experimental conditions. This baseline model has three free parameters. The hierarchy is then extended by systematically increasing the number of free parameters. Each model in the hierarchy is nested within a constrained model posited higher in the branch. So, for example, a more elaborated model is one in which μ varies freely across conditions, while τ and σ are held fixed. When tested in the complete design this model has 6 free parameters (i.e., 4 μs varying across conditions + 1σ + 1τ). To select the best model, one can compare the model with the lower log likelihood when the two models have the same number of parameters, or use the Akaike information criterion (AIC; Akaike, 1983) when the number of parameters is different.

The main analysis consisted of analysis of variance (ANOVA) performed on the best-fitting parameter values derived for each participant, in each condition by the first method. The results of the hierarchical model selection were used to confirm the results of the first method.

Ex-Gaussian parameters in the partial design

Mu. The left panel of Figure 5 presents the values of μ as a function condition. As can be seen, these were comparable across conditions. A *t* test did not reveal a difference between the aligned and misaligned conditions—$t(19) = -0.48$, $d = 0.07$, $p = .68$.

Sigma. The middle panel of Figure 5 presents the values of σ as a function of the partial design conditions. The difference was far from significance—$t(19) = 0.79$, $d = 0.19$, $p = .45$.

Tau. The right panel of Figure 5 presents the values of τ as a function of alignment. A *t* test, $t(19) = 1.89$, $d = 0.24$, $p < .05$, revealed a significant effect, with larger values of

τ in the aligned (252 ms) than in the misaligned (216 ms) conditions, mimicking the exact pattern of RT and error rate in the partial design (Rossion, 2013). These results clearly show that the locus of the partial design composite face effect is exclusively in the τ component.

Table 1 gives the results of the hierarchical model selection. As can be noted, the best-fitting models are models in which only τ varies across distributions, or more elaborated models in which τ varies with other parameters. However, improvements in fit were too small to warrant the more elaborative model—$\chi^2(4)=8$, $p>.05$.[5] So, the winning model is the one with less parameters (only τ) varying across distribution. Note, however, that even when all three parameters could vary, as is the case in the traditional method of fitting, the statistical analyses were significant only for the τ component. Therefore, the results of the two methods are completely in agreement.

Complete design

Mu. The left panel of Figure 6 presents the values of μ as a function of the complete design conditions. As can be seen, these values did not vary across conditions. A two-way ANOVA with alignment and congruency did not reveal effects whatsoever. The effects of alignment, $F(1, 19)=1.94$, $MSE=1{,}742$, $\eta_p^2=0.08$, $p=.17$, congruency, $F(1, 19)=0.13$, $MSE=38$, $\eta_p^2=0.0006$, $p=.72$, and their interaction, $F(1, 19)=0.18$, $MSE=80$, $\eta_p^2=0.00089$, $p=.67$, did not reach significance levels.

Table 1. Summary of the fit statistics of the ex-Gaussian models in the partial design of Experiment 1.

Model	Parameter	Free parameter	–Σ log likelihood	AIC
1	–	3	473	952
2	μ	4	468	944
3	σ	4	454	916
4	τ	4	375	758
5	μ, τ	5	375	560
6	μ, σ	5	423	856
7	τ, σ	5	374	758
8	μ, σ, τ	6	371	754

AIC: Akaike information criterion.
See text for details.

Sigma. The middle panel of Figure 6 presents the values of σ as a function of the complete design conditions. The only significant effect was that of alignment, $F(1, 19)=5.19$, $MSE=4{,}523$, $\eta_p^2=0.20$, $p<.05$, entailing lower values of σ in the aligned (48 ms) than in misaligned (62 ms) faces. The effect of congruency was far from significance ($F<1$), as was their interaction, $F(1, 19)=1.79$, $MSE=1{,}103$, $\eta_p^2=0.08$, $p=.19$.

Tau. The right panel of Figure 6 presents the value of τ as a function of the complete design conditions. A two-way ANOVA with alignment and congruency on values of τ revealed a main effect of alignment, $F(1, 19)=4.41$, $MSE=15{,}338$, $\eta_p^2=0.18$, $p<.05$, such that longer τ values observed in aligned than in misaligned faces. A main effect of congruency, $F(1, 19)=7.69$, $MSE=15{,}338$, $\eta_p^2=0.18$, $p<.05$, entailed shorter τ values in congruent than in incongruent trials. A significant interaction of Alignment × Congruency, $F(1, 19)=7.75$, $MSE=10156$, $\eta_p^2=0.28$, $p<.05$, mimicked the exact pattern observed with RTs and error rates in the complete design measure, namely, a congruency effect on τ values in the aligned condition, $F(1, 19)=10.44$, $MSE=16{,}575$, $\eta_p^2=0.35$, $p<.005$, but not in the misaligned condition, $F(1, 19)=0.32$, $MSE=189$, $\eta_p^2=0.01$, $p=.57$. These results entail that the locus of the composite effect in the complete design is in the τ component, whereas the alignment manipulation affects the σ component.

Table 2 presents the results of the hierarchical model selection. As can be noted, best-fitting models were those in which τ varied. Among these models, the best-fitting model was decided based on the AIC index, which penalised

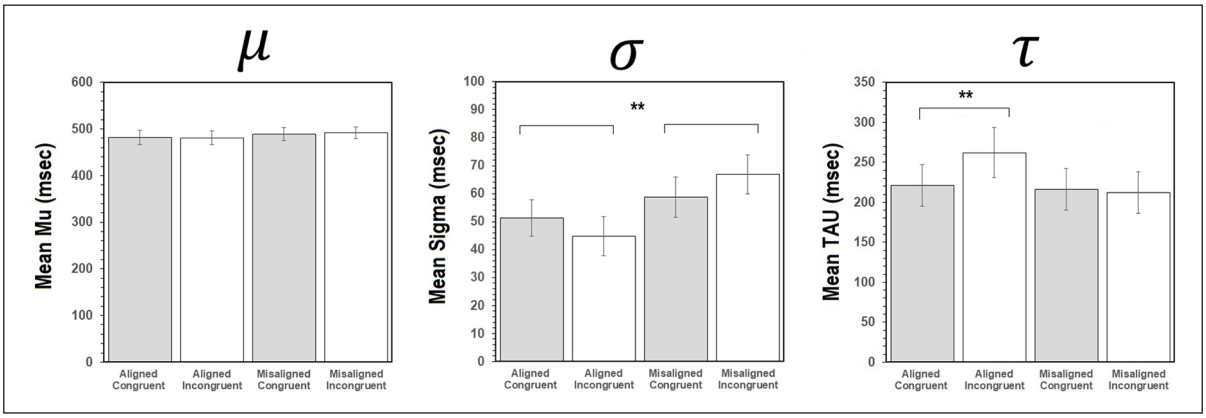


Figure 6. Experiment 1: values of mu, sigma, and tau, as a function of alignment and congruency in the complete design. Error bars are the standard error of the mean.

Table 2. Summary of the fit statistics for the ex-Gaussian models in the complete design of Experiment 1.

Model	Parameter	Free parameter	–Σ log likelihood	AIC
1	–	3	880	1,766
2	μ	6	880	1,772
3	σ	6	860	1,732
4	T	6	708	1,428
5	μ, τ	9	707	1,432
6	μ, σ	9	850	1,718
7	τ, σ	9	706	1,430
8	μ, σ, τ	12	706	1,436

AIC: Akaike information criterion.
See text for details.

additional parameter. The winning model is therefore one in which only τ varies across distributions. This result converges on the finding from the traditional method in which all three parameters are fitted to all distributions.

Discussion

Experiment 1 replicated the classic finding of a composite face effect with unfamiliar faces in both the partial and complete designs. Most importantly, the effects were decomposed into their ex-Gaussian components through the fitting of the ex-Gaussian parameters to the RT distributions. The results were clear-cut; the composite face effect (in both partial and complete designs) selectively affected the exponential τ component of the RT distribution. What can be deduced from this pattern? First, such a pattern increases our belief in the view that partial and complete measure tap into common processes (Fitousi, 2019b). Second, since τ is strongly associated with attentional and working memory processes (Schmiedek et al., 2007) and/or object-based attention processes (Spieler, 2001), the results support attentional- and object-based accounts of the composite face effect. Third, since unique influence on the τ component are tantamount with effect on the tail of the RT distribution, the results entail that the effect emerges at late percentiles rather than early percentiles. This temporal dynamics is inconsistent with the template account (Jacques & Rossion, 2009), which predicts a decrease in the magnitude of the composite face effect with time. It is more consistent with selective attention views (Fitousi, 2015; Richler & Gauthier, 2014), which predict that the features or parts are initially represented a set of independent entities, but progressively interact due to failures of selective attention (see Fitousi, 2019b for discussion).

Experiment 2

One question that begs an answer given the consistent findings of Experiment 1 is whether composite face effects with familiar and unfamiliar faces are governed by the same mechanisms (Burton et al., 2015). The ex-Gaussian analysis allows us to test this question by looking at the fine-grained details of the underlying RT distributions. If we find that the composite effects for familiar and unfamiliar faces are manifested at different ex-Gaussian components, we can infer that they are governed by different mechanisms. Thus, the goal of Experiment 2 was to replicate the original composite face effect with familiar faces (Young et al., 1987) at the level of MRTs, and then to apply the ex-Gaussian analyses to its RT distributions.

Method

Participants. Twenty-six undergraduate students (7 males, 19 females) were recruited from Ariel University's pool of participants (M_{age}=23.34, SD=2.58). They received course credit for their participation. All participants reported normal or corrected-to-normal vision. All participants gave their written consent for participation.

Stimuli. Images of four very famous people: two politicians (Donald Trump and Vladimir Putin) and two actors (George Clooney and Tom Cruise) were used. The faces in the images were front viewing with a smiling expression (but not exposing teeth). The images were transformed into grey scale and cut in an oval shape that kept the faces relatively comparable with respect to size, brightness, and shape (see Figure 7). The four images were cut in half into eight top and bottom parts. All top and bottom parts were aligned with each other to create 16 faces from which 12 were completely new composites, and 4 were the original identities (Trump, Putin, Clooney, and Cruise). In the aligned condition, top and bottom halves were presented exactly one above the other. In the misaligned condition, the top halve was shifted to the middle of bottom half (see Figure 7). In total, there were 32 different stimuli in this experiment. Viewed at a fixed distance of 76 cm, the aligned images subtended 4.4° of visual angle, horizontally, and 2.3° vertically, whereas the misaligned images subtended 6.5° of visual angle, horizontally, and 2.3° vertically.

Procedure and design. Each trial started with a black screen for 500 ms, followed by a face that was presented until response. Aligned and misaligned test composite faces were presented in alternated blocks. Each block consisted of 80 trials with all four possible combinations of study and test faces (same-same, same-different, and different-same) equally presented. Each block was repeated six times with the order of stimuli randomly decided by the computer. Overall, each participant completed 960 experimental trials.

The experimenters verified that participants were familiar with the four identities by asking them to name their faces. Participants were instructed to recognise the top half of the face and ignore the bottom half. They were asked to

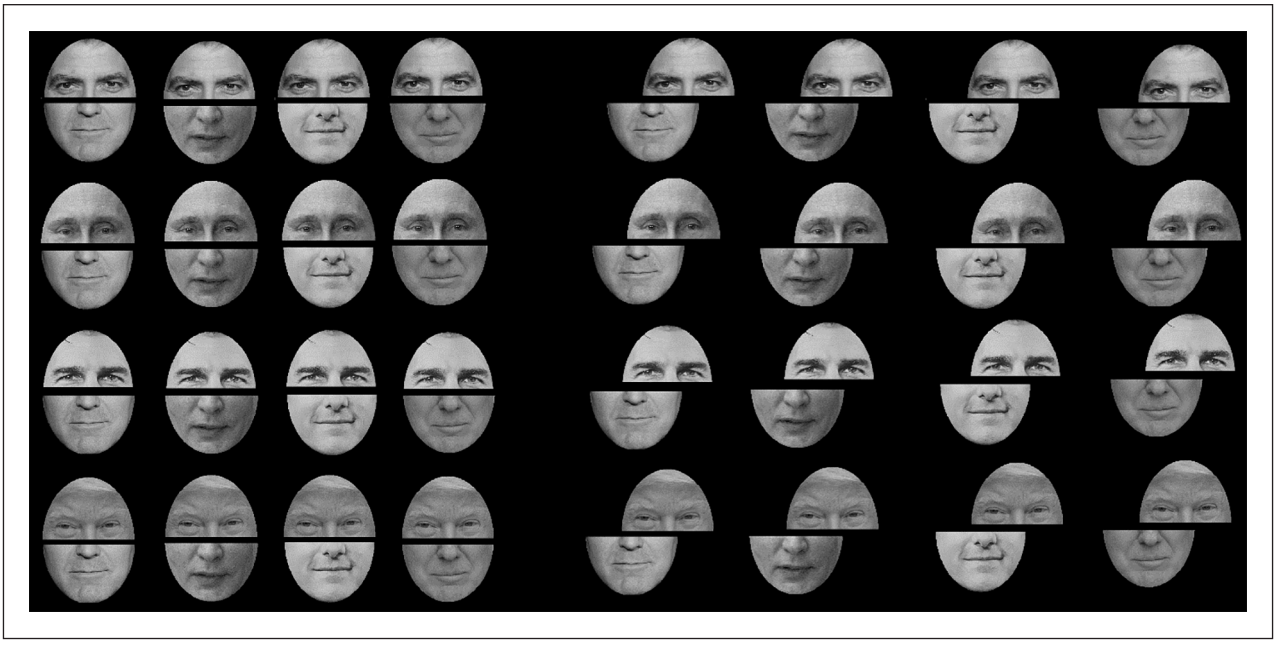


Figure 7. The stimuli set in Experiment 2. The faces were created by composing the top and bottom halves from four famous identities: George Clooney, Vladimir Putin, Tom Cruise, and Donald Trump. All possible combinations (4 × 4) were used in aligned and misaligned configurations. The 16 composites on the left were presented in the "aligned condition" whereas the same 16 composites on the right were presented in the "misaligned condition." The original identities (congruent faces) are located across the diagonals, whereas the novel composites (incongruent faces) are located off diagonals. In the experimental blocks, congruent and incongruent composites appeared in the same frequency.

press a key as fast and as accurate when they identify the top half of the face according to the following assignment: George Clooney ("z"), Vladimir Putin ("x"), Tom Cruise ("m"), and Donald Trump ("n"). The first pair of keys were pressed by the index and middle finger of the left hand, whereas the last pair of keys were pressed by the index and middle finger of the right hand. Response mapping was kept constant for all observers and across all conditions (aligned and misaligned).

Results

RTs either larger than 3,800 ms or smaller than 150 ms were removed from analysis.

Composite effect in the partial design. Figure 8 gives MRTs and mean error rates in the aligned and misaligned conditions for "same" trials only. The partial design effect is a difference in performance measures in the two conditions. The effect was significant for RT, $t(25) = 1.71$, $p < .05$, but not for error rates, $t(25) = -0.15$, $p = .55$.

Composite effect in the complete design. Congruent faces were those in which both the bottom and top halves belong to the same identity (e.g., Clooney), while incongruent faces were those in which the top and bottom halves belong to different identities (e.g., top Clooney, bottom Trump). Figure 9 presents the MRTs and mean error rates as a function of alignment and congruency. ANOVA on MRTs showed that the effect of alignment was not significant, $F(1, 25) = 1.09$, MSE = 25,318, $\eta_p^2 = 0.04$, $p = .30$. The effect of congruency, $F(1, 25) = 20.33$, MSE = 21,315, $\eta_p^2 = 0.44$, $p < .005$, entailed faster RTs for congruent than incongruent faces. Most importantly, evidence for the presence of a composite face effect was provided by a significant Congruency × Alignment interaction, $F(1, 25) = 16.16$, MSE = 14,096, $\eta_p^2 = 0.39$, $p < .001$. Planned comparison showed that a congruency effect was present in the aligned, $t(25) = 5.185$, $p < .005$, but not in the misaligned condition ($t < 1$). These results replicate the standard composite face effect with familiar faces (Young et al., 1987) in the RT data.

A similar analysis was performed on mean error rates. Although insignificant, the error data showed the same trend as the RTs. The main effect of alignment ($F < 1$) and congruency, $F(1, 25) = 1.85$, MSE = 0.0005, $p = .18$, were not significant, and the Congruency × Alignment interaction was on the verge of significance, $F(1, 25) = 3.06$, MSE = 0.0005, $\eta_p^2 = 0.11$, $p = .09$. Due to its insignificance, the interaction term was not further explored.

Fitting the ex-Gaussian distribution

The procedure of fitting ex-Gaussian distribution is comparable to that described in Experiment 1. It consisted of two independent methods. The first looked for the values

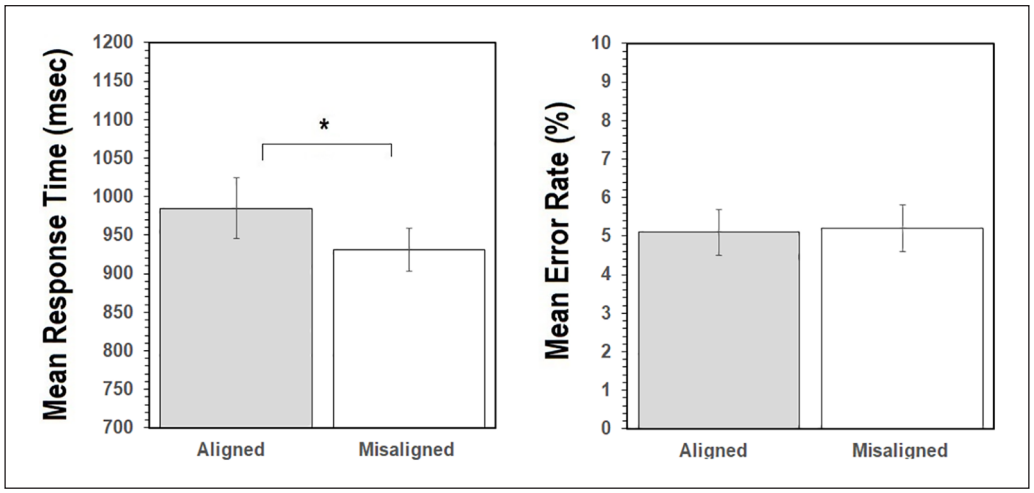


Figure 8. Experiment 2: mean response times (ms) (left panel) and mean error rate (%) in the partial design. Error bars are standard error of the mean.

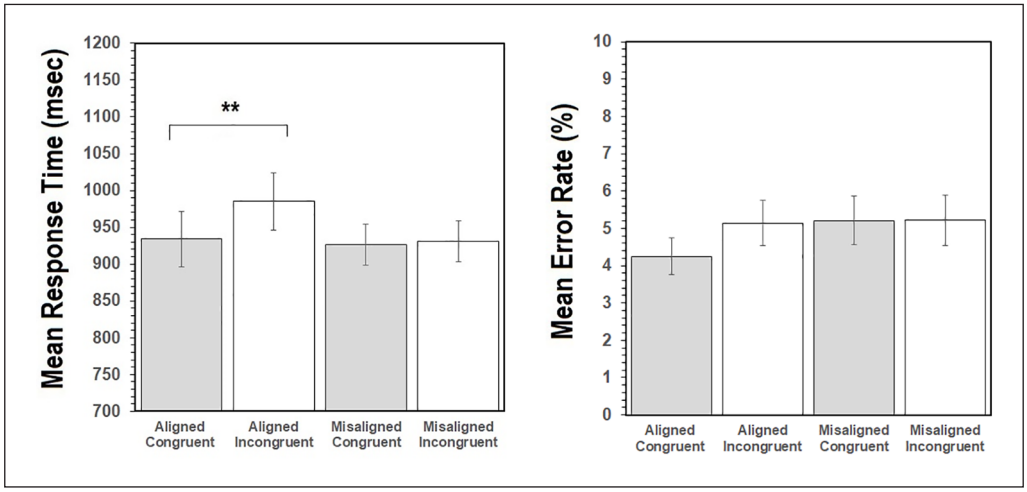


Figure 9. Experiment 2: mean response times (ms) (left panel) in the complete design with familiar faces. Error bars are standard error of the mean.

of the ex-Gaussian parameters: μ, σ, and τ that best predicted the observed data. This procedure was performed for each participant in each condition (104 distributions). The second method consisted of a *hierarchical model selection* (Ashby & Lee, 1991; Fitousi, 2014, 2018) in which models with varying degrees of flexibility are compared (see Experiment 1 for more details). As in Experiment 1, the results from the methods converged on the same conclusions. I report on the best-fitting ex-Gaussian values from the first method. These were supported by the model selection work from the second method.

Ex-Gaussian parameters in the partial design

Mu. The left panel of Figure 10 presents the values of μ as a function condition. As can be seen, these were comparable across conditions. A *t* test revealed no difference between the aligned and misaligned conditions, $t(25) = -0.48$, $p = .69$.

Sigma. The middle panel of Figure 10 presents the values of σ as a function of the partial design conditions. The difference was not significant, $t(25) = 0.81$, $p = .45$.

Tau. The right panel of Figure 10 presents the values of τ as a function of alignment. A *t* test, $t(25) = 1.96$, $p < .05$, revealed a significant effect, with larger values of τ in the aligned conditions, mimicking the exact pattern of RT in the partial design (Rossion, 2013). These results reveal again that the locus of the partial design composite face effect is exclusively in the τ component.

Table 3 presents the results of the model selection procedure for the partial design. Based on the AIC index,

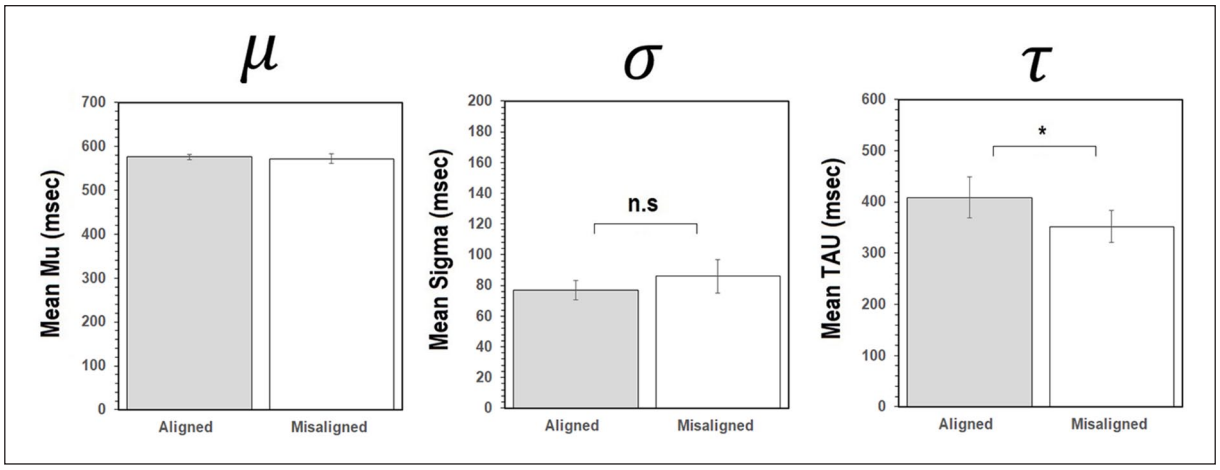


Figure 10. Experiment 2: values of mu, sigma, and tau, in the partial design. Error bars are the standard error of the mean.

Table 3. Summary of the fit statistics of the ex-Gaussian models in the partial design of Experiment 2.

Model	Parameter	Free parameter	–Σ log likelihood	AIC
1	–	3	634	1,274
2	μ	4	529	1,066
3	σ	4	671	1,350
4	τ	4	495	998
5	μ, τ	5	495	1,000
6	μ, σ	5	529	1,068
7	τ, σ	5	495	1,000
8	μ, σ, τ	6	494	1,000

AIC: Akaike information criterion.
Best-fitting model is highlighted. See text for details.

which imposes a penalty on models with additional parameters, the best-fitting model is one in which only τ is varying. This result is completely in agreement with the results of the first method, in which the ex-Gaussian parameters are fitted simultaneously to all the RT distributions.

Ex-Gaussian parameters in the complete design

Mu. The left panel of Figure 11 presents the values of μ as a function of alignment and congruency. A two-way ANOVA with alignment and congruency revealed only an effect of congruency, $F(1, 25)=5.36$, $MSE=5{,}023$, $\eta_p^2=0.17$, $p=.03$. The effect of alignment ($F<1$), and the Alignment × Congruency interaction ($F<1$) were far from significance.

Sigma. The middle panel of Figure 11 presents the values of σ as a function alignment and congruency. None of the effects reached significance level. The effect of alignment was far from significance ($F<1$), the effect of congruency was on the verge of significance, $F(1, 25)=3.69$, $MSE=5{,}416$, $\eta_p^2=0.12$, $p=.07$, and the Alignment × Congruency effect was not significant, $F(1, 25)=1.04$, $MSE=1{,}367$, $\eta_p^2=0.04$, $p=.31$.

Tau. The right panel of Figure 11 presents the value of τ as a function of congruency and alignment. A two-way ANOVA with these factors on values of τ showed that the effects of alignment, $F(1, 25)=1.61$, $MSE=21{,}215$, $\eta_p^2=0.06$, $p=.21$, and congruency ($F<1$) were not significant. Most importantly, a significant interaction of Alignment × Congruency, $F(1, 25)=7.05$, $MSE=14{,}964$, $\eta_p^2=0.22$, $p<.05$, was found. This interaction mimicked the exact composite effect pattern observed with RTs, namely, a congruency effect on τ values in the aligned condition, $t(25)=1.91$, $p<.05$, but not in the misaligned condition, $t(25)=-0.94$, $p=0.82$.

Table 4 presents the results of the model selection procedure. As can be noted, the best-fitting model is one in which only τ is varying. This model exhibited the highest degree of fit in terms of the AIC. These results are in complete agreement with the findings from the first method, in which all three ex-Gaussian parameters were fit simultaneously to the RT distributions. The two methods converge on the conclusion that the composite face effect in the complete design is exclusively manifested at the τ component.

Taken together, the results from Experiment 2 replicate the findings from Experiment 1 with unfamiliar faces. They document manifestation of the composite face effect at the τ component. The signature pattern of an interaction of congruency with alignment was mimicked only in the τ parameter but not in the other ex-Gaussian parameters. These results entail that the locus of familiar and unfamiliar share the same distributional pattern, as they influence the same ex-Gaussian parameter.

Discussion

Experiment 2 replicated the classic composite face effect with familiar faces in a manual response version (Young et al., 1987). The effect was then decomposed into its ex-Gaussian components. The results showed that the

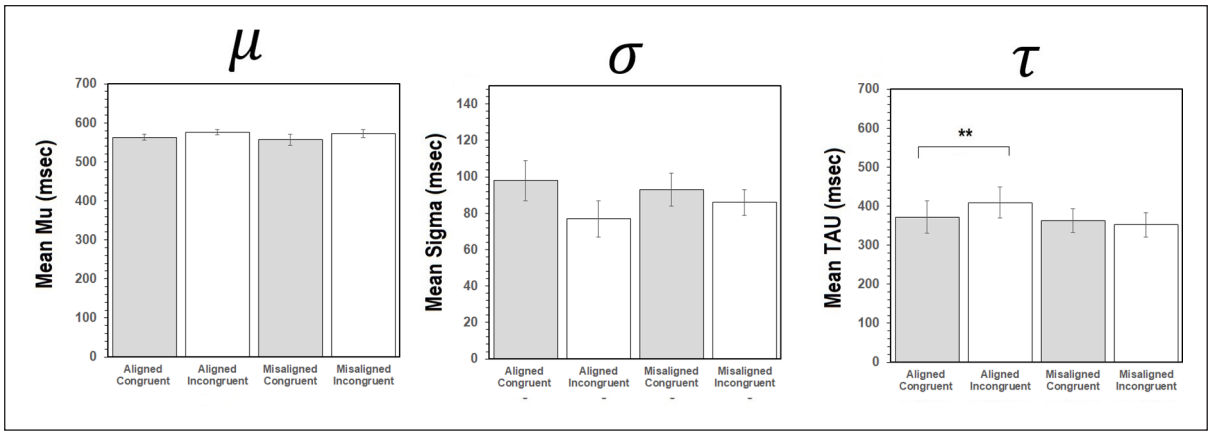


Figure 11. Experiment 2: values of mu, sigma, and tau, as a function of alignment and congruency in the complete design. Error bars are the standard error of the mean.

Table 4. Summary of the fit statistics of the ex-Gaussian models in Experiment 2.

Model	Parameter	Free parameter	–Σ log likelihood	AIC
1	–	3	1,439	2,884
2	μ	6	1,258	2,528
3	σ	6	1,439	2,890
4	τ	6	1,110	2,232
5	μ, τ	9	1,109	2,236
6	μ, σ	9	1,257	2,532
7	τ, σ	9	1,110	2,238
8	μ, σ, τ	12	1,108	2,240

AIC: Akaike information criterion.
Best-fitting model is highlighted. See text for details.

composite face effect is manifested as a unique influence on the exponential τ component of the RT distribution in both partial and complete design measures. This replicates the pattern observed with unfamiliar faces in Experiment 1, lending further support to the view that partial and complete measures tap into common processes (Fitousi, 2019b). This also strengthens the position that familiar and unfamiliar composites are governed by similar (holistic?) processes/constructs (Abudarham et al., 2019). Moreover, since the τ component is associated with attention and working memory processes (Schmiedek et al., 2007) and/or object-based attention processes (Spieler et al., 2001), the current findings support attentional and object-based accounts of the composite face effect (Fitousi, 2015). They also undermine the position that the composite face effect is unrelated to the Stroop effect (Gauthier et al., 2018). Finally, the results also demonstrate that the locus of the composite face effect is at the tail of the RT distribution, which entail stronger effect at late percentiles than early percentiles. This temporal dynamics is inconsistent with the template account (Jacques & Rossion, 2009), which predicts a decrease in the magnitude of the composite face effect with time.

General discussion

The composite face effect (Young et al., 1987) is a central phenomenon in the face recognition literature (Fitousi, 2015; Rossion, 2013). Its popularity can be readily ascribed to the powerful impression it inflicts on viewers, as well as to its robustness across various tasks and stimuli. It is therefore not surprising that the composite face effect has been routinely deployed by face researchers to uncover the mechanism of face recognition under the assumption that it reflects holistic processing (Curby et al., 2013; Hole, 1994; Michel et al., 2006; Richler, Gauthier et al., 2008; Weston & Perfect, 2005). The great majority of the studies with composite faces relied on central tendency measures such as MRTs. However, the mean conceals a great deal of information lurking at the distributional level (Balota & Yap, 2011; Ratcliff, 1978). This study goes well beyond the mean, relating the empirical RT distributions to the ex-Gaussian parameters that generated them (Heathcote et al., 1991). Two experiments were performed with unfamiliar and familiar faces. The experiments yielded highly consistent results, revealing a common pattern in the ex-Gaussian components. First, partial (Rossion, 2013) and the complete (Richler & Gauthier, 2014) composite face effects were found to be associated with a pure influence on the τ parameter. Second, this pattern was found for both familiar and unfamiliar faces.

The composite face effect possibly taps into attention and working memory capacities

This study discovered unique and consistent ex-Gaussian patterns in the RT distributions of the composite face task. The effect consistently emerged only at the exponential component. Capitalising on these regularities, the study was capable of testing novel hypotheses regarding the engagement of constructs and mechanisms that hitherto were difficult to test. At the outset of this work, I have predicted that if the template account (Rossion, 2013) is

correct, and the composite face effect is a purely holistic effect, the composite effect should emerge in the Gaussian components (μ and/or σ), not in the exponential (τ) component. In contrast, if the object-based attention account (Fitousi, 2015) is correct, the composite face effect should emerge at the exponential (τ) component, not in the Gaussian components. The predictions for the selective attention account (Richler & Gauthier, 2014) were not straightforward, as the account commits to an attentional mechanism that is on the one hand "attentional," but on the other hand face-specific and dissociated from those found in classic conflict tasks (e.g., Stroop; Gauthier et al., 2018). In any event, the current results are consistent with the idea that the composite face effect reflects late, controlled, and nonanalytic process. This conclusion is in accordance with the object-based account and (partially) with the selective attention account (Richler & Gauthier, 2014), but not with the template account (Rossion, 2013).

The finding of an exclusive influence on the τ component supports the possible involvement of non-holistic mechanisms such as task conflict, object-based attention, and working memory (Fitousi, 2015, 2016). These constructs have been shown to relate to the τ component (Roelofs, 2012; Schmiedek et al., 2007; Tse et al., 2010; Unsworth & Spillers, 2010). To date, the involvement of such mechanisms in the composite face effect has been denied by all proponents of holistic account (Gauthier et al., 2018; Richler & Gauthier, 2014; Rossion, 2013). Gauthier et al. (2018), for example, argued that neither IQ scores nor Stroop congruency scores are correlated with the composite face effect. However, the contribution of these constructs has been proposed by non-holistic approaches (Fitousi, 2015; see also Cheng et al., 2018). Fitousi (2015) has argued that the composite face effect is a special case of attention-based attention (Duncan, 1984) with face stimuli, and the current results add support in this conjecture. In this sense, faces are not "special," and are subjected to the same cognitive mechanisms that govern other stimuli.

The current results raise the possibility that the composite face task is not a pure measure of holistic processing, but rather a manifestation of more general capabilities, such as working memory. If so, how can one explain the role of working memory in composite face tasks other than the delayed same/different task, such as in the naming task (Young et al., 1987), or a simultaneous-matching task, in which the two faces presented at once (Cassia et al., 2009; Hole, 1994)? A straightforward answer can be given in paradigms that use choice reaction time (CRT). Schmiedek et al. (2007) have argued that CRTs require arbitrary mapping between stimulus and response, and therefore an appropriate selection is mediated by the binding of the correct responses to stimuli in working memory. According to this account, working memory capacities determine the participant's ability to retrieve the response in a fast and efficient fashion. Wilhelm and Oberauer (2006) found that working memory capacities were correlated particularly with CRTs of arbitrary stimulus-response mapping. The current results cannot speak directly to the naming task because this task has not been tested here. But in principle, such a task can potentially involve working memory capabilities indirectly through the known linkage between working memory, intelligence, and speed of processing (Hale & Jansen, 1994). Thus, more intelligent participants are equipped with larger working memory and faster processing speed.

Composite effects with familiar and unfamiliar faces: same or different mechanisms?

It is often found that performance with familiar faces is superior to that with unfamiliar faces, leading researchers to argue that familiar and unfamiliar faces are processed by different mechanisms (Young & Burton, 2017). It has been proposed that familiar face recognition is based on internal facial features, whereas unfamiliar face matching relies on external facial features (Ellis et al., 1979; Kramer et al., 2018). According to another proposal (Jenkins & Burton, 2011), the representation of familiar faces is based on their averaged appearances, whereas the representation of unfamiliar faces relies on image-based information. Burton et al. (2015) have argued that recognition of familiar faces, in contrast to unfamiliar faces, does not depend on configural strategies. Familiar faces can be readily recognised despite configural changes, such as in the case of caricatures (Benson & Perret, 1991). The hypothesis that familiar and unfamiliar faces are processed differently can be tested in the context of the composite face illusion. On the surface, the central tendency measures that capture the composite face illusion may not reveal any difference between familiar and unfamiliar faces, but a closer look at the ex-Gaussian components can potentially do that. The current findings showed that the composite face effect with familiar and unfamiliar faces has been manifested at the same ex-Gaussian component—the τ component. This common pattern increases support in the hypothesis that familiar and unfamiliar faces are governed by the same mechanisms (Abudarham et al., 2019). However, this conclusion should be constrained by the possibility that different mechanisms can still produce similar influence on the τ component.

Studying individual differences in face recognition with the ex-Gaussian components

Many face researchers have recently started to apply an individual-differences approach to probe basic face recognition mechanisms (DeGutis, Mercado, et al., 2013; DeGutis, Wilmer, et al., 2013; Mackenzie et al., 2017). Individual scores in various face and non-face tasks are collected and their correlations are computed (Gauthier et al., 2018). The approach has been harnessed to testing

the validity of the holistic processing idea and its alleged contribution to face recognition abilities. The results of these studies are quite mixed, with some of the studies being supportive of the idea (DeGutis, Mercado, et al., 2013; DeGutis, Wilmer, et al., 2013) and other less so (Gauthier et al., 2018; Konar et al., 2010; Mackenzie et al., 2017; Rezlescu et al., 2017; Wang et al., 2012). One possible reason for the absence of correlations in many of these studies is lack of reliable individual differences in traditional markers of holistic processing (Sunday, Richler, & Gauthier, 2017). One implication of this study is that researchers can use the individual scores of the τ values rather than the MRT to uncover theoretical constructs associated with the composite face effect. The ex-Gauss components can provide a more accurate and reliable measure than the MRTs or accuracy (Saville et al., 2011). Another promising venture for future research is using the τ parameter to study the contribution of working memory capacities to face recognition (Gauthier et al., 2018).

Support in a local-to-global temporal dynamics

Influence on the τ component imply that the effect occurs at late percentiles, whereas effects on the μ component imply a constant magnitude of the effect across time (Balota & Yap, 2011). According to the template hypothesis (Jacques & Rossion, 2009; Sergent, 1986), faces are initially encoded and represented as an integrated whole, rather than a collection of independent features, and then progressively become more refined and detailed. If this account is correct, then the composite face effect should decrease rather than increase with time (see Fitousi, 2019b). Hence, exclusive effect on the τ component is consistent with an increase in the magnitude of the effect as RTs get slower (Balota & Yap, 2011). This implies that according to the template account (Jacques & Rossion, 2009), the composite face effect should not manifest at the τ component. In contrast, the object-based attention (Fitousi, 2015) account predicts a late influence of the global form.

Image variability and the composite face effect

Jenkins et al. (2011) have recently highlighted a built-in limitation in virtually all face studies (including the current one). They noted that by presenting a single image per identity, face studies ignore within-person variation in appearance. This is of course, on the face it, an inappropriate practice, since ecologically speaking, personal identity is conveyed by multiple images, which are subjected to considerable image variability. Jenkins et al. (2011) showed that viewers who are unfamiliar with a certain personal identity perceive its images as belonging to several identities, while viewers who are familiar with this identity performed perfectly. This finding has important consequence to the study of faces in general, and interesting implications for the study of the composite face effect. Consider a composite face experiment with unfamiliar faces that introduces image variability to the top relevant half. In the "same" condition, the "same" identity is coming from a "different" image. This of course may disrupt the phenomenal illusion, because now it is the "same" identity, but the viewer experiences it as a "different" identity. The upshot is that the essence of the illusion is not rooted in the processing of facial identities, but in the processing of images. In this sense, one can replace the face images with those of other objects. This invalidates a holistic account, supporting an object-based approach. Future work should investigate this issue further.

Conclusion

The theoretical weakness of the holistic accounts, along with the possibility for the involvement of more general attentional and working memory mechanisms in the processing of composite faces, invite a conceptual change in the research on faces. Students of faces should not limit their theories to modular "face-specific" processes (e.g., holism), but open to the possibility that faces engage general-purpose cognitive mechanisms, such as attention and working memory. This possibility is highly plausible, since the evolution of the visual system has not been confined to faces, and because the perception of faces engages other visual and attentional areas in the brain beyond the fusiform face area (FFA). One reason that research in face recognition progress slowly (Burton, 2013) is because face recognition is considered a separate ability from other general cognitive processes such as attention and memory. Approaches that endorse a strict modularity might be inappropriate since they ignore the fact that specific behavioural phenomena are generated by more general mechanisms. In this sense, the notion of holistic processing as a unique characteristic of faces has limited rather than expanded our ability to understand face recognition. Burton (2013) has put this point succinctly: ". . . the field has come to rely on poorly-specified notions, such as 'configural processing,' which have hindered rather than helped" (p. 1468). This study expands the scope of theoretical possibilities by relating face processing to more general mechanisms such as attention and working memory using fine-grained methods of RT distributions analyses.

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Notes

1. Young et al. (1987) measured the composite effect as a difference in performance between aligned and misaligned (incongruent) composite faces. This measure is comparable to that computed in the partial design. Young et al. provided a similar "configural" account to that proposed by Rossion (2013).
2. Schwarz and Miller (2012) examined various forms of delta plots and have shown that increasing, decreasing, or zero slope delta plots cannot be generated by the same process model. Others (Balota & Yap, 2011) have shown that whether the delta plot is increasing, decreasing or zero slope depends uniquely on specific ex-Gaussian parameters. So, for example, an increasing delta plot entails a change in the μ parameter. A corollary that follows is that if two effects surface at different ex-Gaussian parameters, they are likely producing different delta plots governed by different processing mechanisms.
3. Note that the Navon figures allow integration from two components that inhere a hierarchical space, and the flanker stimuli enhance integration from two spatially separated components. Hence, the presence of effects on τ in the integrated Stroop paradigm supports this conjecture.
4. I thank a reviewer for acknowledging these potential problems and suggesting the alternative inferential test.
5. The chi-square is constructed such that the chi-square equals twice the difference in log likelihood (LL) between the two models ($2 \times$ [LL Model 1 − LL Model 2]), and the degrees of freedom equal the difference in parameters between the two models (*N* parameter Model 1 − *N* parameter Model 2). See Fitousi (2014, 2018) and Ashby and Gott (1988).

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